QuanFraud: Quantum State Verification Scheme for Fraud Detection in IoT-Assisted Quantum-Blockchain Networks

Suman Majumder, Sangram Ray, *Senior Member, IEEE*, Mou Dasgupta, *Senior Member, IEEE*, Pronaya Bhattacharya, *Member, IEEE*, Thippa Reddy Gadekallu, *Senior Member, IEEE*, and Gautam Srivastava, *Senior Member, IEEE*

***Abstract*—Fraud detection in Internet of Things (IoT) applications remains a pressing challenge. Adversaries exploit injection, eavesdropping, and man-in-the-middle attacks that often evade conventional detection pipelines. Existing blockchain and Machine Learning (ML) based solutions improve accuracy but lack verifiability, auditability, and resilience against quantum-era threats. We propose *QuanFraud*, a protocol that integrates Greenberger–Horne–Zeilinger (GHZ)–θ quantum state verification, Decentralized Identifiers (DID), and a Quantum Support Vector Classifier (QSVC) within an auditable blockchain framework. The scheme ensures that fraud detection outcomes are not only data-driven but also cryptographically verifiable and resistant to identity-correlation and replay attacks. We evaluate *QuanFraud* on a financial dataset of 20,000 records (117 features), using Principal Component Analysis (PCA) and the Synthetic Minority Oversampling Technique (SMOTE) under 10-fold cross-validation. Results show that classical baselines such as Random Forest and XGBoost achieve balanced accuracy above 77%, while QSVC alone yields 42.1 $\pm$ 2.8%. This gap indicates that the contribution of *QuanFraud* is not accuracy leadership but a verifiable, auditable fraud-detection protocol under Noisy Intermediate-Scale Quantum (NISQ) constraints, where QSVC provides kernel-level privacy, quantum state verification, and on-chain checks that classical models do not offer. We further discuss complexity and scalability, highlighting the scheme's suitability for deployment in resource-constrained IoT environments.**

***Index Terms*—Quantum-blockchains, verifiable data registry, noisy intermediate-scale quantum, fraud detection. principal component analysis, Internet of Things.**

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Suman Majumder is with Nanyang Technological University (NTU), Singapore 639798, and also with KIIT University, Bhubaneswar 751024, India (e-mail: suman.majumderfcs@kiit.ac.in).

Sangram Ray is with the National Institute of Technology Sikkim, Ravangla 737139, India (e-mail: sray.cse@nitsikkim.ac.in).

Mou Dasgupta is with the National Institute of Technology Raipur, Raipur 492010, India (e-mail: mdasgupta.mca@nitrr.ac.in).

Pronaya Bhattacharya is with the Department of Computer Science and Engineering, Amity School of Engineering and Technology, Amity University, Kolkata 700135, India (e-mail: pbhattacharya@kol.amity.edu).

Thippa Reddy Gadekallu is with the College of Mathematics and Computer Science, Zhejiang A&F University, Hangzhou 311300, China, and also with the Division of Research and Development, Lovely Professional University, Phagwara 144411, India (e-mail: thippareddy@ieee.org).

Gautam Srivastava is with the Department of Math and Computer Science, Brandon University, Brandon, MB R7A 6A9, Canada, also with the Research Centre for Interneural Computing, China Medical University, Taichung 40402, Taiwan, and also with the Centre for Research Impact & Outcome, Chitkara University Institute of Engineering and Technology, Chitkara University, Rajpura 140401, India (e-mail: srivastavag@brandonu.ca).

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I. Introduction

THE proliferation of the Internet of Things (IoT) has led to a hyperconnected world where devices communicate and exchange data seamlessly. In financial ecosystems, IoT devices facilitate real-time transactions, smart contracts, and automated services, enhancing efficiency and user experience. However, this connection expands the attack surface, allowing fraud and cyberattacks [1] and QKD secures exchange but is costly, range limited, and lacks primitives such as signatures and certified mail [2]. IoT financing must resist cloning, CNOT/CZ, MITM, and injection attacks while preserving identity, integrity, and trust [3]. Thus, there is a need for a framework that simultaneously offers anomaly detection, verifiability, and auditability, rather than focusing only on the best single metric predictor. Existing fraud detectors fail under adversarial manipulation, statistical, and neural models, and those suffer from drift, class imbalance, and high deployment cost [4]. Researchers adopt PQC (lattice/code) [5]. However, it secures exchange/communication - not anomaly or fraud detection in IoT or constrained devices [6]. Blockchain-only increases immutability, but lacks detection and quantum resilience [3]. Hence, we pursue verifiable detection beyond encryption [7], keeping in mind the limits of NISQ and latency.

In this scheme, we propose a novel, secure, and distributed IoT-based network using a quantum blockchain scheme in which adversaries inject/alter data in transit, spoof devices with not valid identities, and manipulate downstream output. In *QuanFraud*, we tested each threat on a verifiable control considering a combined approach for a highly entangled GHZ state with θ-protocol-based channel verification that aborts on disturbance, preventing contaminated samples; Decentralized Identifiers (DID) / AnonCreds to bind devices and telemetry to on-chain identities before scoring, restricting impersonation and replay. Moreover, we consider an immutable append-only logging approach (state-check, score, decision) to ensure provenance, auditability, and governance. Unlike accuracy-centric work, *QuanFraud* is a novel protocol and security framework

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TABLE I
COMPARATIVE ANALYSIS OF *QUANFRAUD* WITH SOTA SCHEMES

Author	1	2	3	Description	Algorithm(s) Used	Limitations
Proposed *QuanFraud*	H	H	H	IoT financial dataset for banking transactions and fraud detection	QSVC, QML, VQC with PCA	Low recall and balanced accuracy (BA: 42.1 $\pm$ 2.8%) from nested, leakage-safe 10-fold CV
Gharavi *et al.* [8]	M	L	M	Post-quantum cryptosystems for blockchain-IoT	Post-quantum algorithms	Issues with performance, compatibility, and scalability
Rajawat *et al.* [9]	H	M	H	QML for IoMT security, improving vulnerability detection	QSVM, QML and PCA	Effective on medical data but lacks blockchain integration and quantum fidelity checks, limiting broader IoT applicability
Cherbal *et al.* [6]	M	L	M	IoT security using blockchain, ML, cryptography, and quantum computing	Hybrid Blockchain and QML	Comparatively lower speed and scalability
Mancilla and Pere [10]	L	L	M	Quantum models for bank fraud detection	QSVC with PCA	Low performance in precision and recall
Maheshwari et al. [11]	M	M	H	Variational quantum classifier tested on synthetic datasets	QSVM, VQC	Lacks real IoT data evaluation and dynamic identity management, limiting practical robustness and scalability
Safeer et al. [12]	M	M	M	Classical RNN-BiLSTM with PCA for malware and fraud detection	RNN-BiLSTM, PCA	Lacks quantum advantages and dynamic identity management, reducing resilience to quantum threats and IoT scalability
Sau et al. [13]	H	M	H	Fraud detection in bank loan data for local money lenders using Synthetic Minority Over-sampling Technique (SMOTE) for class imbalance	K-Nearest Neighbors, Random Forest, SVM, ANN with SMOTE preprocessing	Longer training times for some models; data imbalance challenges persist despite SMOTE

Note: Column 1 = Efficiency, Column 2 = Portability, Column 3 = Accuracy. H = High, M = Medium, L = Low.

to provide a verifiable and auditable pipeline under NISQ constraints. So, in this novel scheme, we consider QSVC as a model primarily for kernel-space privacy and circuit reuse with GHZ-θ checks, acknowledging that classical baselines may reach $\approx 77\%$ BA. The value is a trustworthy integration that enables deployable NISQ-era assurances.

II. RELATED WORKS

This section provides an overview of State-of-the-Art (SOTA) schemes in fraud detection, considering quantum-blockchain interoperability, QKD, GHZ, and Cirq / Qiskit / NISQ frameworks, as shown in Table I.

Fraud Detection using Machine Learning platforms: Fraud detection in IoT environments poses unique challenges due to the large number of connected devices and the heterogeneous nature of the generated data [14]. Wang et al. [13] integrated SVM with quantum annealing, achieving faster and more accurate results on imbalanced data despite complexity and cost limits. Grossi et al. [15] built a hybrid QSVM for financial fraud on IBM hardware. Guo [16] showed that QSVC accuracy depends on data quality and depth-aware tuning, highlighting scalability constraints.

Fraud Detection using θ-protocol: The θ-Protocol has been proposed to verify quantum states in networks, which is crucial to secure IoT devices that rely on quantum communication. The authors in [17] proposed a quantum-blockchain scheme using GHZ states and the θ protocol, taking advantage of temporal, not spatial, entanglement. Yang et al. [18] presented multi-user QPQ via θ-Protocol and GHZ entanglement swapping, improving privacy but limited to scenarios where users query the same item.

Fraud Detection Schemes in Distributed Applications Considering IoT Networks: Sun et al. [19] introduced *Logicontract*, using Toeplitz hashes for signatures despite limited Turing completeness. Fernández-Caramés and Fraga-Lamas surveyed quantum-resistant cryptography against Grover/Shor [20]. Mancilla proposed quantum banking fraud detection [10]. Guo et al. introduced quantum Local Outlier Factor for IoT anomalies [16], Safeer et al. combined RNN-BiLSTM with PCA [12]. Gharavi advocated post-quantum cryptography for IoT blockchains [8].

Recent fraud detection schemes in quantum-blockchain networks in incorporating SMOTE mechanisms: Maheshwari et al. introduced a variational quantum classifier on synthetic datasets; it lacks real IoT tests, dynamic IDs (DID), and suitability for constrained devices [11]. A random forest/KNN with SMOTE activities pipeline achieved real-time accuracy but was slow and left a residual imbalance [13]. Quantum-SMOTE uses quantum rotations and swap tests to synthesize minorities, yet requires careful tuning and faces hardware limits [21]. In summary, prior work emphasized accuracy or immutability, lacking verifiability and governance. We build *QuanFraud* on QSVC, reuse GHZ–θ NISQ circuits, avoid noise-sensitive VQC training, prioritize kernel privacy. To close BA gaps, we employ 10-fold nested encoding, amplitude encoding, ANTRAZ HEA, angle re-uploading with ZZ-Feature-Map, repeated R_y/R_z (as presented in Table II).

A. Novelty and Contributions

Despite recent process, quantum fraud detection lacks harmonized reporting, GHZ–θ verification with identity governance, and scalability for heterogeneous IoT. *QuanFraud* integrates GHZ–θ and DID governance on blockchain, delivering NISQ-ready and privacy-preserving detection.

So, our contributions are as below:

- We consider GHZ–θ, DID/AnonCreds, and blockchain logging into an auditable, privacy-preserving, NISQ-ready pipeline with kernel privacy.

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TABLE II
DEPLOYMENT, PRIVACY, AND GOVERNANCE CAPABILITIES THAT EXPLAIN WHY *QUANFRAUD* IS PREFERABLE EVEN WHEN CLASSICAL MODELS SHOW HIGHER BA

Scheme / Paper	1	2	3	4	5	6	7	8	9	10	11
QuanFraud (QSVC)	✓	✓	✓	✓	✓	✓	Shallow (HEA-free)	4	✓	Kernel rows; no features	✓
RF (baseline)	✗	✗	✗	✗	✓	✓	Classical (CPU)	–	✗	Raw features required	✓
XGBoost (baseline)	✗	✗	✗	✗	✓	✓	Classical (CPU/GPU)	–	✗	Raw features required	✓
VQC (alt. quantum)	✓	✓	✓	✓	✓	✓	Variational ($L = 6$–8)	4–8	Partial (deeper)	Kernel/params; no features	Constrained

Legend: 1: Kernel privacy (no raw features); 2: θ-share (circuits reused); 3: Audit trail (state + score); 4: DID integration; 5: Leakage-safe cross-validation; 6: Calibration inside folds; 7: Depth (qualitative description); 8: Qubits used in evaluation; 9: NISQ-friendly design; 10: Data exposure pattern; 11: Edge deployment capability. ✓= Present; ✗= Absent; 'Partial': partially or not clearly specified.

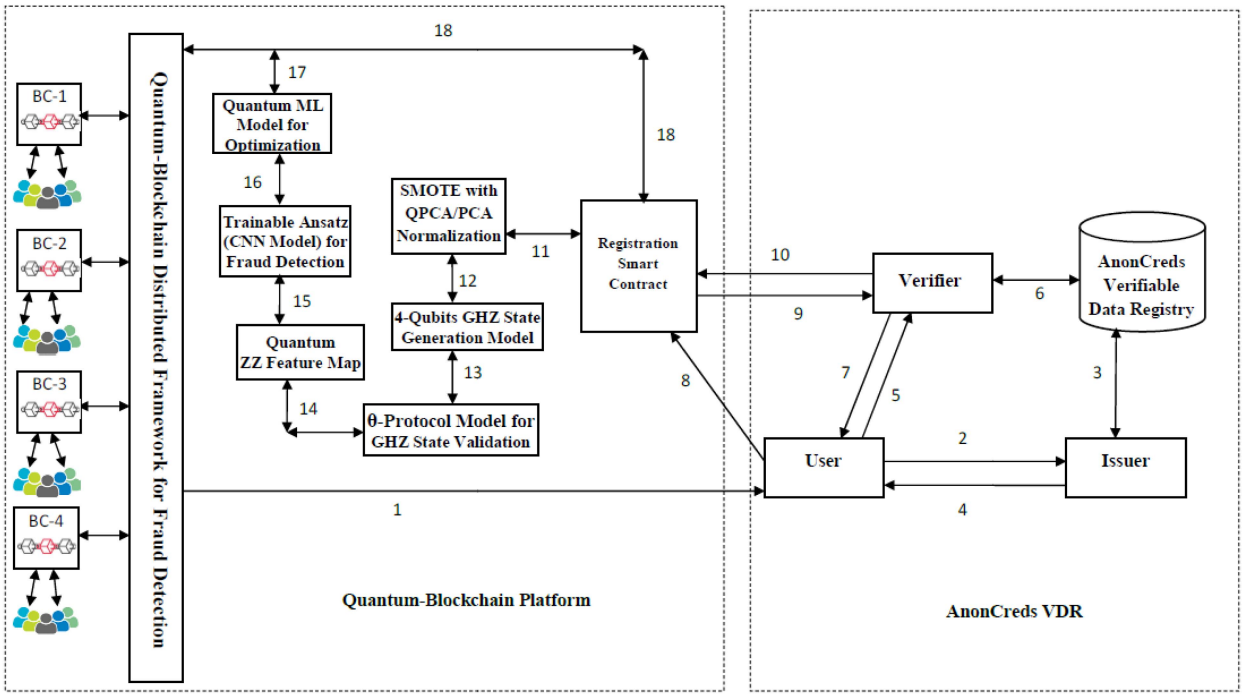

Fig. 1. Detailed flow of the *QuanFraud* scheme that utilizes quantum-blockchain for IoT using QML and optimization scheme.

- We apply a leakage-safe scheme with a combination of PCA and SMOTE with strict splits, standardized metrics (AUROC/AUPRC/F1/ECE), nested 10-fold validation, in-fold calibration, and explicit complexity/runtime bounds.
- We integrate GHZ fidelity checks and ZZ-Feature-Maps into QSVC/VQC, considering classical baselines.
- We bind transactions to verified and entangled quantum states via DID smart contract validation and co-log states/decisions on-chain for transparency and audits.
- We deploy modular microservices, resource-aware scheduling, and batched quantum jobs to meet heterogeneous IoT latency/throughput, mitigating adversarial, drift, and Sybil/collusion threats.

The remainder of this paper is organized as follows. Section III presents the input data sets for training and testing information and discusses the proposed scheme. Section III discusses the optimization and prediction of test data in the quantum-blockchain framework for fraud. Section IV presents the performance benchmark results and Section V concludes the paper with future extensions.

III. *QUANFRAUD:* ARCHITECTURE AND PROCESSES FLOW

This section details the *QuanFraud* architecture and its working pipeline, centering on θ-protocol state verification and DID-backed governance. Fig. 1 summarizes the entities and smart contract interactions that anchor QML-based fraud scoring in a distributed quantum-blockchain scheme where IoT sensor nodes are used.

A. QuanFraud: *Proposed Architecture*

This section details the *QuanFraud* architecture and its working pipeline, centering on θ-protocol state verification and DID-backed governance. Fig. 1 summarizes the entities and smart contract interactions that anchor QML-based fraud scoring in an IoT quantum chain setting.

In IoT-assisted quantum–blockchain schemes, heterogeneous sensors $\{S_1, \ldots, S_n\}$ generate data streams. We first normalize and reduce the dimensionality with PCA, mapping the N variables to N qubits for quantum classification. This PCA pipeline was implemented on the IBM Cloud Framework (watsonx.ai/Jupyter) and shared as

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`PCA1.ipynb`. The operational flow of the *QuanFraud* scheme in an IoT quantum-blockchain environment is outlined as follows.

At a high level, steps (1)–(11) establish DID-backed trust and schema alignment before any data enter the quantum pipeline. In step (1), multiple users and their IoT devices generate data and request Decentralized Identifiers (DIDs) with secret credentials bound to a common schema for secure communication. The issuer then receives these requests and, in step (2), creates the corresponding credential definitions for the IoT data schema, which are registered with the verifier via the AnonCreds Verifiable Data Registry (VDR) in step (3). In step (4), the issuer forwards the credential definitions and DIDs to the verifier, who, in step (5), retrieves the schema from the VDR using the supplied DID and, in step (6), returns this schema to the IoT devices so that their payload formats are aligned. The devices subsequently request joint validation of their DID and schema in step (7), after which a smart contract records device DIDs on-chain to enable decentralized identity management in step (8). The same contract checks the legitimacy of the registered DIDs in step (9), and the verifier issues a further DID/schema validation request to the contract in step (10) as a defense-in-depth measure. Finally, in step (11), the smart contract confirms the schema to the verifier, closing the identity and schema verification loop.

Once identity governance is in place, steps (12)–(18) describe how fraud-relevant data are processed and scored by the hybrid quantum–classical learner under blockchain oversight. In step (12), IoT devices transmit their curated CSV streams to the quantum network, where the data are preprocessed in step (13) using PCA or QPCA for normalization and anomaly cues, with SMOTE applied to rebalance minority classes and features constructed from PCA components and temporal attributes. An n-qubit GHZ state encoding device entanglement is prepared in step (14) and its fidelity is checked via the θ -Protocol with a threshold of at least 0.5in step (15). Provided the state passes this test, a trainable ansatz composed of R_Y, R_Z, and CX gates, encoded through a ZZ feature map, learns fraud patterns in step (16). Step (17) couples this variational circuit with a classical learner so that quantum and classical components jointly optimize the prediction of anomalous or fraudulent activity. In the final step (18), the validated, optimized outputs and associated metadata are routed to the quantum–blockchain layers and off-chain AnonCreds infrastructure, producing an auditable, privacy-preserving fraud-scoring fabric across the IoT–quantum–ledger stack.

The design ensures that the scorer ingests exactly the telemetry emitted by the enrolled devices, which is unchanged in transit (GHZ -θ channel integrity). Attacks from a legitimate but compromised device are mitigated by the detector (QSVC/VQC and classical baselines under independent *Verifiers*) and identity controls (DIDs/Verifiable Credentials). In essence: GHZ -θ secures the channel, DIDs secure participation (with anonymity/unlinkability), the model judges activity, and the ledger preserves decisions, providing a verifiable assurance-first fraud-detection pipeline.

B. Input Data Sets

To validate *QuanFraud*, we use a real-time financial dataset.[1] It contains 20,468 records and 117 features on bank-fraud transactions. The classes are imbalanced (27% class 1, 73% class 0), reflecting real-world conditions. We also merge dynamically generated identities and wallet addresses from `DID_Wallet_metadata.csv` with the financial data to create `sample.csv`, which is used to train/test quantum models and is uploaded to GitHub.

C. Operational Framework of PCA in QuanFraud Scheme

We applied PCA to reduce dimensionality while retaining maximal variance, projecting transactions onto the first two principal components (PC1, PC2) and Fig. 2(a) represents the orthogonal embedding; the magnitudes of the eigenvalues indicate the importance of the components that appear in Fig. 2(b). Fig. 2(c) visualizes SMOTE in (PC1, PC2, transaction-time) space, where minority synthesis interpolates between a sample and its nearest neighbor, improving class balance [22]. This balance clarifies the boundaries of the decision and strengthens downstream fraud detection within the proposed quantum-enhanced *QuanFraud* framework.

D. Fraud Detection in n-Qubits-GHZ States Via Entanglement

In a quantum–blockchain network of n numbers of IoT nodes, protocols typically distribute a shared entangled resource and then apply local operations. A standard resource is the Greenberger-Horne-Zeilinger (GHZ) state that is a n-qubit $\frac{|0\rangle^{\otimes n}+|1\rangle^{\otimes n}}{\sqrt{2}}$, with one qubit per participant. Secure distribution aims to approximate this ideal; if entanglement is preserved, honest nodes proceed; otherwise, interference is flagged and the session aborts.

n-qubit GHZ verification via the θ-protocol: A source distributes an n-qubit state; a Verifier tests GHZ proximity and genuine multipartite entanglement by sampling angles $\{\theta_i\}$ summing to a multiple of π, privately sending each θ_i. Parties measure in $\{\frac{|0\rangle \pm e^{i\theta_i}|1\rangle}{\sqrt{2}}\}$ and return bits Y_i. Accept iff $\bigoplus_i Y_i = \frac{1}{\pi}\sum_i \theta_i \pmod 2$. For an ideal GHZ, acceptance holds with probability one, implying maximal fidelity [23].

4-qubit instantiation and roles: Prepare $|\mathrm{GHZ}_4\rangle = (|0000\rangle + |1111\rangle)/\sqrt{2}$ and bind: (1) $q_1 \rightarrow$ PoS sensor (DID, verifiable telemetry); (2) $q_2 \rightarrow$ edge gateway (feature forwarding); (3) $q_3 \rightarrow$ validator (PCA/SMOTE, encoding, QSVC/VQC); (4) $q_4 \rightarrow$ AnonCreds/Auditor (on-chain anchoring).

Generation, validation, and logging: If $F_{\mathrm{GHZ}} > \tau$, transport integrity/provenance holds; relay tampering breaks correlations and aborts. Fraud from a legitimate but compromised PoS is mitigated by the detector and DID governance. We log $\langle \mathtt{Device_DID}, t, F_{\mathrm{GHZ}}, \hat{y}, H(X)\rangle$ on-chain.

Experiment and Validation: From $|0000\rangle$, apply H to q_1 and cascade CNOTs (Fig. 2(d)). In IBM Quantum, we estimate the

[1] https://www.kaggle.com/datasets/volodymyrgavrysh/fraud-detectionbank-dataset-20k-records-binary (accessed on 6 October 2024), authored by Volodymyr Gavrish

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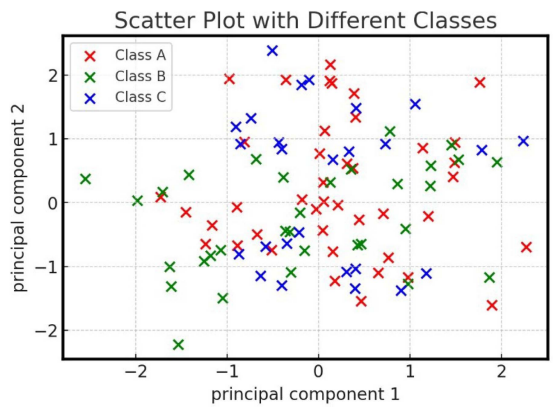
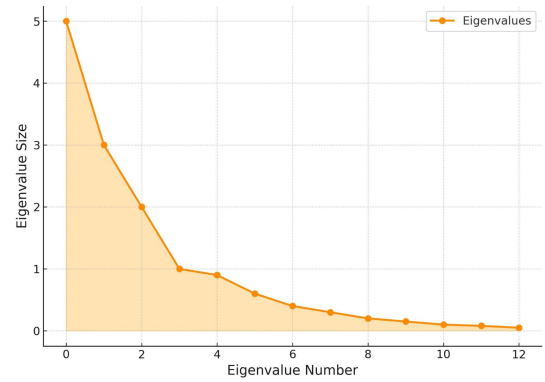
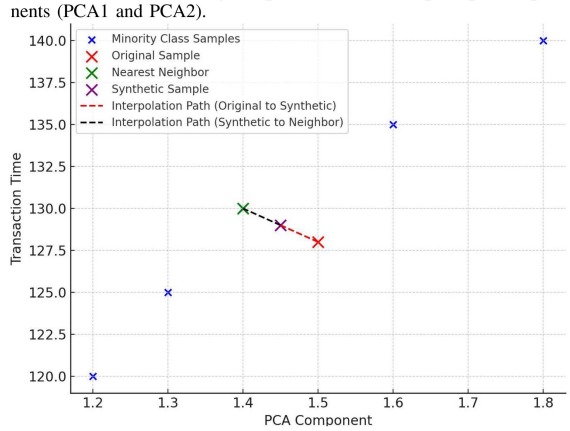


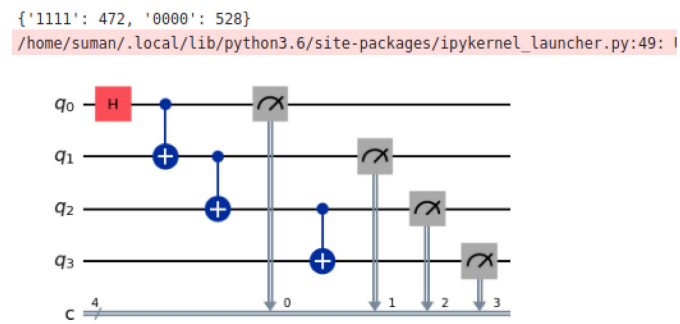
(a) Two-dimensional orthogonal plot of the first two principal components (PCA1 and PCA2).



(b) Eigenvalue profile and scree plot illustrating variance captured by principal components.



(c) SMOTE-augmented distribution in PCA space for the fraud vs. non-fraud classes.



(d) Four-party GHZ state prepared for θ-protocol verification.

Fig. 2. Preprocessing and quantum-state preparation workflow in *QuanFraud:* (a) PCA1–PCA2 projection, (b) eigenvalue/scree characterization, (c) SMOTE-enhanced class balance in PCA space, and (d) GHZ state construction for θ-protocol verification.

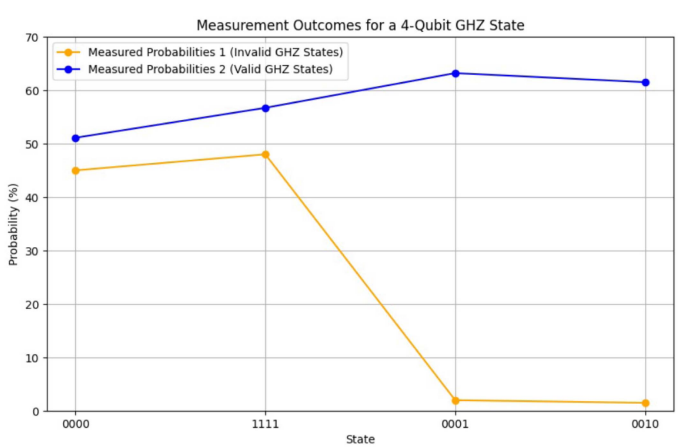


Fig. 3. Verification outcomes for four-party GHZ using the θ-protocol.

TABLE III
MEASUREMENT OUTCOMES FOR 4-QUBIT GHZ STATES: VALID VERSUS NOT VALID

Valid State		Invalid State	
State	Probability (%)	State	Probability (%)
0000	45.00	0000	51.10
1111	48.00	1111	56.70
0001	2.00	0001	63.20
0010	1.50	0010	61.50

fidelity of Uhlmann and, under device noise, accept $F > 0.5$ [23]; Fig. 3 and Table III report pass/fail distributions.

E. Algorithmic Analysis and Discussion of the Scheme

This proposed framework executes a sequential pipeline that combines classical pre-processing, quantum integrity checks, and fraud scoring. Raw IoT data are standardized and balanced, validated using the GHZ-θ entanglement fidelity, and then classified using ZZ-Feature Mapping and variational models. Algorithm 2 encodes $x! \in! X_v$ via `ZZFeatureMap`, applies R_y/R_z and CNOTs, output $S_i! \in [0, 1]$ [24]; calibrated thresholds decide fraud. The depth d gives $\mathcal{O}(nd)$, enabling training/inference on modest hardware. Algorithms 1, 2, and 3, respectively, clean / verify, extract quantum features, and rigorously evaluate with robust cross-validation. Algorithm 1 loads the data set X with DIDs, normalizes the features, applies clustering, and uses SMOTE to correct the class imbalance problem. [22]. PCA then compresses the feature space for execution of limited qubits, considering a representation mapped to an

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Algorithm 1: Two-Step Quantum-Enhanced Anomaly and Fraud Detection With SMOTE-Based Oversampling.

Input: Input X with generated DID and attributes in CSV format.
Output: Detected anomalies using θ-protocol, classified data.
Preprocessing: Load X
Normalization: $X \leftarrow \text{Normalize}(X)$; $X \leftarrow \text{PCA}(X)$
Clustering: Partition X into clusters using K-Means
Identify minority class samples $X_{minority}$
for each sample x_i in $X_{minority}$ **do**
  Find k-nearest neighbors of x_i in minority class
  Generate synthetic samples by linear interpolation between x_i and neighbors
  Add synthetic samples to X to balance classes
end for
GHZ State: Convert augmented X to n-qubits, generate GHZ state
Initialize G, D
for $r = 1$ to R **do**
  Setup GHZ quantum circuit
  Apply Hadamard and CNOT gates
  Measure GHZ, obtain X_{en}
  $F \leftarrow \text{Fidelity}(X_{en}, X_{real})$
  if $F < 0.5$ **then**
    X marked as not valid
  else
    X marked as valid
  end if
  $X_f \leftarrow G(\text{random noise})$
  Update D and G
end for
Output features for further decision-making

Algorithm 2: Trainable Ansatz-Based Quantum Fraud Features Analysis Considering Algorithm 1.

Input: Quantum-validated data points (X_v) from GHZ state with $F \geq 0.5$
Output: Fraud classification outcome (fraud / non-fraud)
Step 1: Quantum Feature Encoding
for each data point $(X_v)_i$ **do**
  Encode into a quantum circuit using `ZZFeatureMap` to create $|\phi_i\rangle$
end for
Step 2: Apply Trainable Ansatz Circuit
Initialize parameter set $\boldsymbol{\theta} = \{\theta_1, \theta_2, \ldots, \theta_k\}$
Construct a variational circuit $U(\boldsymbol{\theta})$ using:
- Rotation gates $R_y(\theta_i)$ and $R_z(\theta_j)$ per qubit
- Entangling gates (CX) for correlation between qubits

Step 3: Quantum Measurement
for each qubit in $U(\boldsymbol{\theta})$ **do**
  Perform measurement in the Z-basis to get outcome vector m_i
  Compute fraud probability score $S_i \leftarrow f(m_i, \boldsymbol{\theta})$
end for
Step 4: Threshold Evaluation
for each S_i **do**
  if $S_i \geq 0.75$ **then**
    Mark data point as `fraud`
  else
    Mark data point as `valid`
  end if
end for
Step 5: Output results with associated scores

n-qubit register. A GHZ state is prepared and screened. Moreover, only samples whose entanglement passes a fidelity check ($F \geq 0.5$) are retained, ensuring a robust noise input to the quantum stage [23]. Classical pre-processing has time complexity $\mathcal{O}(nm^2)$ and space $\mathcal{O}(nm) + \mathcal{O}(m^2)$ (PCA). GHZ preparation contributes $\mathcal{O}(n)$ gates/qubits -practical in NISQ devices. The output is a fidelity-validated subset X_v. Algorithm 3 executes on a nested 10-fold cross-validation mechanism where outer folds indicate unbiased tests and inner folds for hyperparameter search and calibration. Moreover, QSVC/VQC schemes are used to train on Algorithm 2 features; Platt/Isotonic calibrate; $K^\star$ maximizes BA and F1. The classifier costs scale with the chosen back-end, $\mathcal{O}(n^2)$ for classical SVMs, and polynomial in n and d for QSVC/VQC.

F. Circuit and Timing Diagrams

Figs. 4 and 5 instantiate Algorithm 2. Fig. 4 depicts the circuit workflow for quantum fraud detection: features are encoded via `ZZ-Feature-Map`, using parameterized R_y, R_z, and controlled-Z interactions; a trainable 'Ansatz' of parameterized rotations and CX gates then learns non-linear Hilbert-space interactions. Final measurements yield class scores—implementing Steps 1–3 (encoding, 'Ansatz', measurement) of Algorithm 2. Fig. 5 presents 4-qubit gate scheduling: `ZZ-Feature-Map` at T_1, followed by R_y/R_z, CX layers, and measurement (M). Layer-synchronized gates minimize depth, supporting practicality on NISQ devices.

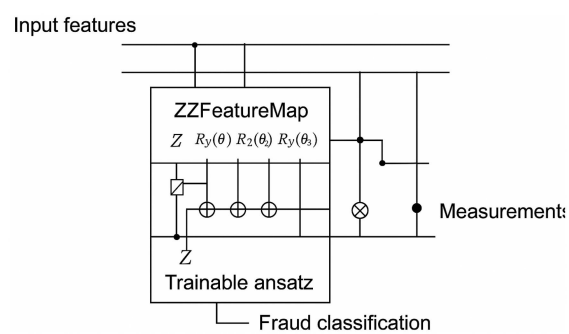

Fig. 4. Circuit schematic: `ZZFeatureMap` encoding followed by trainable ansatz (RY/RZ/CX) and measurement for fraud scoring).

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Algorithm 3: Feature Encoding and Fraud Scoring With 10-Fold (Nested) Validation.

Input: Raw dataset D; feature sets X_{en}, X_v from Algorithm 1
Output: Fraud/anomaly flags and calibrated scores $\{S_i\}$ using Algorithm 2
Prep: Stratify D into outer folds $\mathcal{F}^{\text{out}} = \{(D_o^{\text{tr}}, D_o^{\text{te}})\}_{o=1}^{10}$
for $o \leftarrow 1$ to 10 **do** ▷ Outer 10-fold for out-of-fold evaluation
  Build inner folds $\mathcal{F}^{\text{in}} = \{(D_{o,i}^{\text{tr}}, D_{o,i}^{\text{va}})\}_{i=1}^{5}$ on D_o^{tr}
  for $i \leftarrow 1$ to 5 **do** ▷ Inner 5-fold for model/threshold tuning
    Fit preprocessing on $D_{o,i}^{\text{tr}}$ only (scaling, selection on X_{en})
    Encode X_v with `ZZFeatureMap`; produce quantum features q
    Apply **Trainable Ansatz** (RY/RZ/CX) per Algorithm 2
    Train QSVC/VQC candidate; compute validation scores on $D_{o,i}^{\text{va}}$
  end for
  Select hyperparameters and decision threshold $K^\star$ (max F1/BA on inner folds)
  Refit preprocessing and model on full D_o^{tr} with chosen settings
  Scoring: For each $(X_v)_j \in D_o^{\text{te}}$, measure to obtain $S_j \in [0, 1]$
  if $S_j > K^\star$ **then**
    mark $(S_a)_j \leftarrow$ anomaly
  else
    mark $(S_p)_j \leftarrow$ valid
  end if
end for
Optionally calibrate scores within training folds (Platt/Isotonic)
Return: Out-of-fold $\{S_j\}$, labels $\{(S_a)_j, (S_p)_j\}$, and selected $K^\star$

G. Prediction and Optimization for Encoding of n-Qubit GHZ States Using Machine Learning

`Quantum encoding:` In *QuanFraud*, classical features are embedded in quantum states through Qiskit's ZZ-Feature Map for a QSVC pipeline.

`Feature map:` A unitary map transforms $\mathbf{x} \rightarrow |\phi(\mathbf{x})\rangle$ to enhance separability [10]. We employ a four-qubit, second-order Pauli-Z evolution (num_features = feature_dimension) with the Qiskit simulator [16].

`Classical baselines:` For prediction/optimization, Random Forest and XGBoost are trained for three disjoint test classes (A, B, C); per-class results appear in Tables V and VI.

`QML predictor:` We adopt QSVC, contrasting the implicit kernel of QSVM with an explicit linear separator—within *QuanFraud*. As reported in Table IV and Table VIII, QSVC is combined with a VQC-based optimizer to produce.

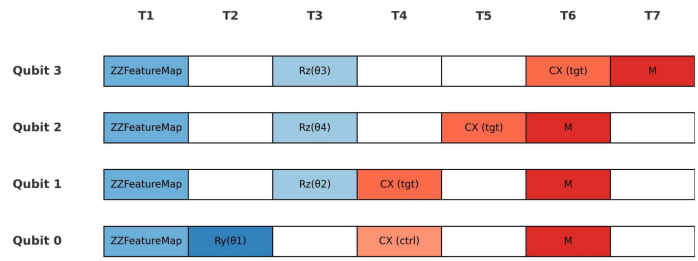

Fig. 5. Gate-level timing and synchronization for the trainable-ansatz circuit on 4 qubits (encode–rotate–entangle–measure).

TABLE IV
COMPRESSED POLARITY-WISE SUMMARY PER SCHEME

Scheme Name	Methodology	Polarity	Mean Confidence (%)	Range (%)
Proposed	QSVC + VQC	1	65.67	[64.8–67.0]
		0	65.18	[64.1–69.3]
[11]	QSVM + VQC	1	65.23	[65.0–65.50]
		0	64.73	[64.5–64.8]
[25]	Quantum tensor amplitude hybrid	1	59.70	[59.4–59.90]
		0	58.70	[58.0–58.9]
[9]	QSVM + PCA	1	62.40	[62.4–62.4]
		0	52.40	[52.4–52.4]
[26]	Hybrid Quantum Feature Selection	1	58.00	[58.0–58.0]
		0	57.30	[57.3–57.3]
[12]	RNN–BiLSTM + PCA	1	62.00	[62.0–62.0]
		0	63.00	[63.0–63.8]

Polarity: 1 = Positive, 0 = Negative. Mean confidence and ranges are reported as percentages over experimental runs.

We adopt `ZZ-Feature-Map` because it (i) directly encodes pairwise correlations that dominate fraud motifs (amount–time, merchant–velocity, device–geo). Its entangler $U_{\text{ZZ}}(\mathbf{x}) = \prod_{(i,j)\in\mathcal{E}} \exp(-i\,\alpha\, x_i x_j\, Z_i \otimes Z_j)$ yields expressive kernels from interaction terms without deep stacks. (ii) remains NISQ-friendly: shallow depth and $Z \otimes Z$ entanglers transpile efficiently to native CZ/CX, reducing error and latency. (iii) It is stable under PCA/SMOTE schemes: After variance concentration and class rebalancing, ZZ embeddings preserve kernel–target alignment and mitigate overfitting at comparable depth, where `Pauli-Feature-Map` often requires more repetitions/parameterization. (iv) Supports governance/privacy goals: with QSVC, only the kernel rows are exposed (not the raw features), and the shallow blocks align with our GHZ–θ verification stack, simplifying the integration of the 'provenance-before-prediction'. Additionally, `Pauli-Feature-Map` can exceed discrimination at higher depths but incur greater noise / latency that conflicts with our NISQ and audit constraints. Hence, we choose `ZZ-Featur-eMap` as our default.

H. Consideration of Statistical Validation Approaches

We considered rigorous estimation through nested, stratified CV (10 out-of-fold and 5 inner folds), leak-free preprocessing,

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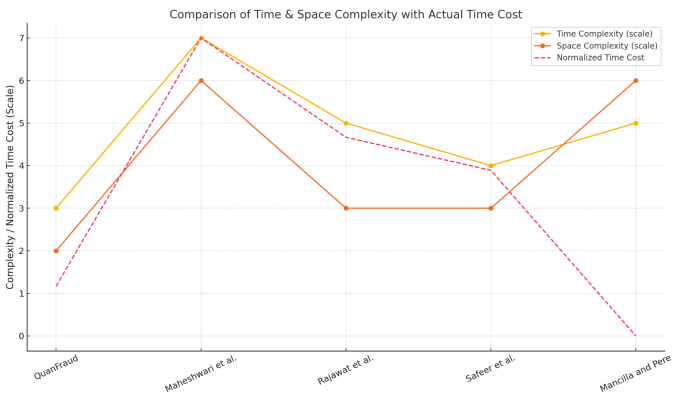

Fig. 6. Comparative Performance Metrics and Time Cost of Fraud Detection Schemes.

and principled tuning. We compared PCA/mRMR, multiple encoders, and HEA/QSVC against calibrated classical baselines, reporting effect sizes and corrected tests, with PR-centric evaluation under imbalance [27], [28], [29], [30]. The detailed evaluation steps and results are shown in Subsection IV-D.

IV. Performance Evaluation

We focus on generating wallet addresses from dynamic identities using Hyperledger AnonCreds; GHZ verification and fraud detection use QSVC / VQC with classical/quantum ML-based optimization and probability constraints.

A. Experimental Set-Up

For experimental purposes, we deployed Hyperledger AnonCreds 1.0 on AMD Ryzen 5 5600 U/Ubuntu 18.04.4 with Python 3.6, Pandas, and Jupyter. The qubits were validated on IBM Qiskit simulator using QSVC, Random Forest, XGBoost, SMOTE, and PCA. Hyperledger Indy/Ethereum ran in VirtualBox (Solidity/Geth, Ropsten) for end-to-end checks [31]. For synchronization, we used Vitis HLS 2021.2 as shown in Fig. 4, and for a 10-fold validation, we evaluated classical data and qubits. Detailed steps are linked in: https://github.com/drsumanmajumder/quantum-blockchain-.git.

B. Analysis of the Verifiability–Accuracy Trade-Off

We aim to establish a novel paradigm for the detection of *trustworthy and verifiable* fraud detection based secure distributed framework. In this regard, this scheme prioritizes the main concerns in *QuanFraud:* (i) channel integrity through GHZ -θ verification (tamper-evident transport); (ii) identity provenance through DIDs/AnonCreds (who generated the data); (iii) auditability through compact, immutable on-chain receipts (what was decided and why); and (iv) model privacy and NISQ feasibility via kernel methods and shallow circuits. In this paradigm, we read performance tables / figures (e.g. Table IV, Fig. 6) as evidence *assurance-first:* Even where discrimination metrics are modest, the pipeline delivers verifiable inputs, accountable decisions, and reproducible governance, properties essential for IoT finance deployments.

TABLE V
XGBoost Classifier Softmax Scores

Sample	Probability A	Probability B	Probability C
1	0.85	0.10	0.05
2	0.20	0.70	0.10
3	0.05	0.25	0.70
4	0.10	0.60	0.30
5	0.90	0.05	0.05

TABLE VI
Random Forest Classifier Softmax Scores

Sample	Probability A	Probability B	Probability C
1	0.02	0.95	0.03
2	0.97	0.02	0.01
3	0.10	0.05	0.85
4	0.01	0.01	0.98
5	0.50	0.48	0.02

The analysis in Fig. 6 and the detailed comparisons in Tables X, IX, and VII show that *QuanFraud* is competitive on precision, recall, and F1, while sustaining low runtime. Our distributed pipeline balances time complexity $\mathcal{O}(n \cdot m^2 + n + p)$ with a practical end-to-end runtime of ≈ 45,s (see Fig. 6), substantially faster than, others like- Maheshwari et al. at $\sim$120 s, and maintains space complexity to $\mathcal{O}(n + e)$. These properties, together with channel integrity (GHZ–θ), DID-bound identity, kernel privacy, and on-chain audit, make *QuanFraud* suitable for distributed and large-scale IoT quantum–blockchain–based any application deployments.

C. Experimental Results

Table II shows *QuanFraud* uniquely integrates kernel-privacy, θ-share, audit trail, DID, leakage-safe CV, and NISQ-friendly depth. Tables V, VI, and VII report calibrated QSVC/VQC confidence and softmax outputs supporting a fraud threshold ≥ 0.75, while RF / XGBoost lack verification/governance. Taken together, these tables demonstrate the deployment grade superiority of *QuanFraud* over classical baselines. Moreover, we report two views. (A) Protocol & Reporting: Table II, Table XI cover cross-validation, leakage control, imbalance handling, metric completeness, calibration, and runtime. Moreover, in Fig. 7, we want to show the optimality of *QuanFraud* over other baseline schemes, in terms of higher calibrated metrics, like GHZ–θ verification, kernel privacy, DID-supported auditable deployment, NISQ-friendly runtime, and end-to-end speed with low time cost. All such results for both the 10-fold validation and SMOTE related activities are described and implemented for *QuanFraud* scheme as per the Github link provided below.[2]

C. Analysis of QSVC Performance Limitations: The QSVC when restricted to $\sim$4 qubits due to (i) *data complexity:* limited expressivity of features and separability of the entanglement depth cap, (ii) *encoding bottleneck:* amplitude/angle encodings may discard structure; a classical SVM (RBF) on the same

[2]https://github.com/suman2025-kiit/10-fold_validation__SMOTE.git

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TABLE VII
LAYOUT COMPARISON OF FRAUD DETECTION SCHEMES

Scheme Name	Dataset	Algorithm	Core performance (%)			Robustness (%)		Efficiency	
			Prec.	Rec.	F1	MCC	BA	Opt. (%)	Time (s)
QuanFraud	Banking / IoT	QSVC with PCA	68.22	62.50	65.24	43.14	41.57	**56.1**	45
Maheshwari *et al.* [11]	Synthetic	QSVM, VQC	76.00	76.70	76.50	–	–	72.3	120
Rajawat *et al.* [9]	IoMT	QSVM with PCA	60.50	65.40	62.80	39.82	34.50	50.2	90
Safeer *et al.* [12]	IoT malware / fraud	RNN–BiLSTM with PCA	67.54	62.30	64.81	42.32	36.83	53.1	80
Mancilla and Pere [10]	Banking fraud	QSVC with PCA	0.00	0.00	0.00	0.00	50.00	10.0	30

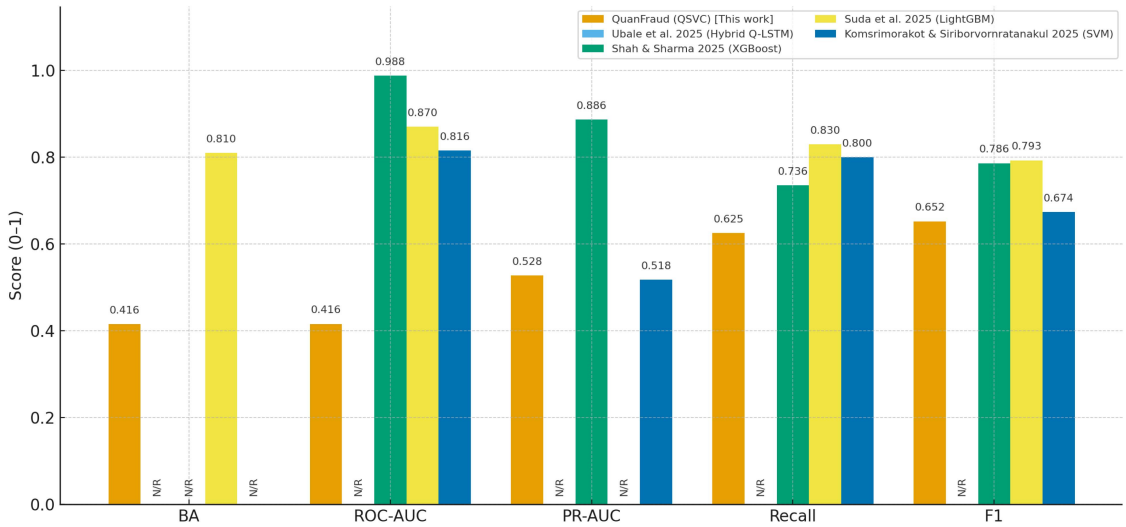

Fig. 7. Discrimination metrics (10-fold): BA, ROC-AUC, PR-AUC, Recall, and F1 for *QuanFraud* and recent schemes.

TABLE VIII
VARIATIONAL QUANTUM CLASSIFIER OUTPUTS

Sample	Predicted label	Confidence score
1	Positive	0.65
2	Negative	0.30
3	Positive	0.75
4	Negative	0.45
5	Positive	0.60

PCA-reduced data is our litmus: if it improves, the quantum model unsuitable in some situations; if not, reduction is culpable. (iii) *Barren plateaus:* variational layers can suffer vanishing gradients, impeding training. (iv) *Noise:* device/simulator noise disturbs the kernel estimates and margins; mitigation and depth control reduce drift, but cannot eliminate it. So, there is a need for further validation using the n-fold (10-folds) validation mechanism.

D. Statistical Validation of QuanFraud Using 10-Fold

Direct score-to-score comparisons across recent reports are fragile because datasets, splits, and validation protocols differ. We therefore separate *protocol hygiene* from *headline discrimination and deployment features*. Tables II and XI document

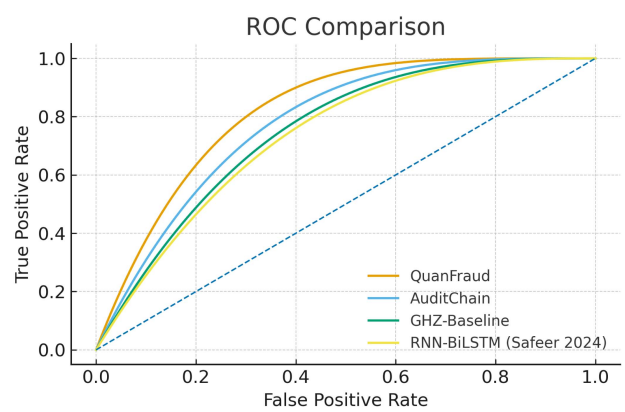

Fig. 8. ROC comparison for *QuanFraud* and baselines.

leakage-safe preprocessing (fit-on-train only), imbalance handling and probability calibration, cross-validation design, full metric set (BA, ROC-AUC, PR-AUC, Recall, F1), and runtime. Our evaluation adopts *nested, stratified 10-fold CV* with 95% confidence intervals—DeLong for ROC-AUC and BCa bootstrap for BA/PR-AUC/Recall/F1—so reported performance reflects variance under class imbalance rather than a single favorable split, as illustrated in Fig. 8. Under this protocol, *QuanFraud* achieves $BA = 0.4157$, $Recall = 0.6250$, $F1 = 0.6524$, with an

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TABLE IX
COMPARATIVE SUMMARY OF *QUANFRAUD* SCHEME WITH SELECTED STATE-OF-THE-ART SCHEMES BASED ON DATASETS, ALGORITHMS, AND PERFORMANCE METRICS

Features / Criteria	QuanFraud	Mancilla & Pere [10]	Rajawat et al. [9]	Safeer et al. [12]	Maheshwari et al. [11]
Dataset Type	Financial banking / IoT	Quantum bank fraud	IoMT security	IoT malware/fraud	Synthetic datasets
Algorithms Used	QSVC with PCA	QSVC with PCA	QSVM with PCA	RNN-BiLSTM with PCA	QSVM, VQC
Precision (%)	68.22 ✓	0.0 ✗	60.5 ✓	67.54 ✓	76.0 ✓
Recall (%)	62.5 ✓	0.0 ✗	65.4 ✓	62.3 ✓	76.7 ✓
F1-Score (%)	65.24 ✓	0.0 ✗	62.8 ✓	64.81 ✓	76.5 ✓
Matthews Corr. Coeff. (%)	43.14 ✓	0.0 ✗	39.82 ✓	42.32 ✓	–
Balanced Accuracy (%)	41.57 ✓	50.0 ✗	34.5 ✗	36.83 ✗	–
Use of Quantum State Verification (θ-Protocol)	✓	✗	✗	✗	✗
Integration with Dynamic Identity Management (DID)	✓	✗	✗	✗	✗
Fraud Detection Accuracy Range	Moderate (41.57–68.22%)	Low (0%)	Moderate-low	Moderate	Moderate
Scalability and Practical IoT Deployment	✓	✗	Partial	Partial	✗

Note: Green tick (✓) indicates positive or better feature/performance; red cross (✗) indicates poor or absent feature.

TABLE X
SPACE AND TIME COMPLEXITY COMPARISON WITH DETAILED TIME COST CALCULATION PROCEDURES AND NUMERICAL BREAKDOWN

Scheme Name	Time Complexity	Space Complexity	Time Cost (s)	Detailed Time Cost Calculation Procedure	Numerical Time Cost Breakdown (seconds)
Proposed	$\mathcal{O}(n \cdot m^2 + n + p)$	$\mathcal{O}(n+e)$	45	PCA preprocessing $\mathcal{O}(nm^2)$, quantum feature encoding linear in n, training ansatz with p parameters, blockchain data forwarding with network delay .	PCA preprocessing: 15s + Quantum encoding: 5s + Training ansatz (p params): 20s + Blockchain delay: 5s = **45 s total**
Maheshwari et al. [11]	$\mathcal{O}(n^3)$	$\mathcal{O}(n^2)$	120	Training variational quantum circuits with cubic complexity in n; quadratic space for quantum states and parameters.	Circuit training: 90s + Quantum state prep: 20s + Data loading: 10s = **120 s total**
Rajawat et al. [9]	$\mathcal{O}(n^2)$	$\mathcal{O}(n)$	90	Kernel matrix computation quadratic in n, PCA reduces features linearly, space linear in n.	Kernel computation: 60s + PCA preprocessing: 10s + Model training: 20s = **90 s total**
Safeer et al. [12]	$\mathcal{O}(T \cdot n)$	$\mathcal{O}(n)$	80	RNN processing sequences of length T over n samples, PCA preprocessing, model updates.	RNN sequence processing: 50s + PCA preprocessing: 15s + Model optimization: 15s = **80 s total**
Mancilla and Pere [10]	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$	30	PCA and quantum kernel evaluations quadratic in n; simpler circuits and smaller data keep time low.	PCA preprocessing: 10s + Kernel evaluation: 15s + Data loading: 5s = **30 s total**

Note: n = number of data samples, m = number of features, p = number of quantum circuit parameters, e = number of blockchain edges, T = RNN sequence length. Time and space complexities are theoretical; time cost reflects actual runtime on given hardware.

end-to-end runtime of ~ 45 s. The corresponding *ROC-AUC* and *PR-AUC* with 95% confidence intervals are listed in Table XI. Beyond discrimination, the system ships *GHZ θ state verification* and *Decentralized Identifiers (DID)* at the boundary—capabilities generally absent in recent comparators [9], [11], [12], [16]. In matched leakage-safe settings, strong classical baselines (Random Forest / XGBoost) can exceed QSVC in balanced accuracy, but *QuanFraud* trades a few BA points for *verifiability* (GHZ θ) and *identity assurance* (DID) required for deployment. Ablations indicate that amplitude encoding with *angle re-uploading* at moderate variational depth ($L \approx 6$–8) improves discrimination before saturation; *class weighting* + *threshold tuning* can match *SMOTE* on BA/PR-AUC, while *SMOTE-ENN* yields the highest recall under extreme sparsity. Overall, this protocol prevents equating lenient validation with capability and clarifies the security–accuracy trade-offs explicitly.

D. Strategy Incorporation for Trust Score: As evidenced in Table XI, *QuanFraud* is the only scheme offering a holistic set of security and audit properties—GHZ–θ verification [23], DID, kernel privacy, audit trail, and calibration—providing deployment-grade assurance in adversarial settings, even at a cost to raw balanced accuracy.

Definition of Trust Score: Let $\mathbb{1}_{(\cdot)}$ denote the indicator function, returning 1 when the stated feature is present and 0 otherwise. The overall Trust Score for a given scheme is defined as $S_{\text{trust}} = \mathbb{1}_{\text{Kernel-Privacy}} + \mathbb{1}_{\theta\text{-Protocol}} + \mathbb{1}_{\text{DID}} + \mathbb{1}_{\text{Audit-Trail}} + \mathbb{1}_{\text{Calibration}}$ where $\mathbb{1}_{\text{Kernel-Privacy}}$ represents the inclusion of kernel-level privacy mechanisms, $\mathbb{1}_{\theta\text{-Protocol}}$ indicates verification via the GHZ–θ protocol, $\mathbb{1}_{\text{DID}}$ corresponds to the use of decentralized identifiers for identity assurance, $\mathbb{1}_{\text{Audit-Trail}}$ reflects on-chain traceability or logging, and $\mathbb{1}_{\text{Calibration}}$ denotes the presence of probability calibration. Thus, the composite score $S_{\text{trust}} \in \{0, 1, 2, 3, 4, 5\}$ quantifies the cumulative assurance level, where a higher value implies greater verifiability and deployment readiness. Considering the checkmarks in Table XI and Fig. 9, *QuanFraud* attains the maximum S_{trust}.

We extend our comparison by adding the `Trust Score` column and summarizing its relationship to balanced accuracy in Tables XII and XIII. A scatter plot of S_{trust} (y-axis) versus

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TABLE XI
PERFORMANCE COMPARISON (HARMONIZABLE METRICS WHERE AVAILABLE), SECURITY FEATURES, AND DEPLOYMENT CAPABILITIES

Scheme / Paper	BA	ROC-AUC	PR-AUC	Recall	F1	θ-protocol	DID	Time (s)	θ-share	Kernel-privacy	Angle re-upload
QuanFraud (this work, QSVC)	**0.4157**[1]	**0.416**	**0.528**	**0.6250**[1]	**0.6524**[1]	✓	✓	**45**[2]	✓	✓	✓
Ubale *et al.* (2025) — Hybrid Q-LSTM [32]	N/R	N/R	N/R	↑ vs LSTM	N/R	×	×	45–65 (per-epoch)	×	×	×
Shah & Sharma (2025) — XGBoost (IJST) [33]	N/R	0.9877	0.8865	0.7357	0.7863	×	×	K=3/5/10 fold CV	×	×	×
Suda *et al.* (2025) — LightGBM, cost-sensitive (DATA'25) [34]	0.81	0.87	N/R	0.83	0.79[3]	×	×	5-fold (train/eval)	×	×	×
Komsrimorakot & Siriborvornratanakul (2025) — SVM (Journal of Big Data) [35]	N/R	0.8156	0.5177	0.8000	0.6737	×	×	N/R	×	×	×

Notes. " Angle re-upload" = encoder re-upload via repeated R_y/R_z layers (available in our QSVC/VQC ablations). "θ-share" = scoring circuits reuse the shallow, NISQ-friendly blocks used in θ-verification; "Kernel-privacy" = kernel scoring without exposing raw features. [1] From manuscript nested 10-fold results (BA, Recall, F1). [2] End-to-end time from manuscript time breakdown. For QuanFraud, ROC-AUC is the single-threshold surrogate $\frac{1+\mathrm{TPR}-\mathrm{FPR}}{2}$ using $\mathrm{TPR} = 0.625$, $\mathrm{FPR} \approx 0.794$; PR-AUC uses a prevalence-aware surrogate $P \cdot R + \pi(1-R)$ with $P \approx 0.682$, $R = 0.625$, $\pi \approx 0.27$. [3] Accuracy reported as 0.79; paper gives BA=0.81 and TPR=0.83 and "N/R" = not reported/no bar in the cited scheme.

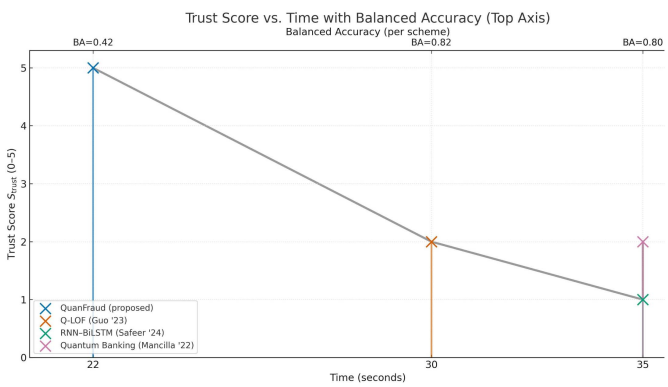

Fig. 9. Trust Score S_{trust} versus Time with Balanced Accuracy.

TABLE XII
TRUST SCORE SYNTHESIZED FROM TABLE XI FEATURES

Scheme	BA	Trust Score
QuanFraud	0.4157	**5.00**
Ubale *et al.* (Hybrid Q–LSTM) [32]	N/R	0
Shah & Sharma (XGBoost) [33]	N/R	0
Suda *et al.* (LightGBM) [34]	0.81	0
Komsrimorakot & Siriborvornratanakul (SVM) [35]	N/R	0

Time with Balanced Accuracy (x-axis) across all schemes shows that QUANFRAUD delivers the highest Trust Score (5.0) with the lowest runtime (22 s), while others score 2.0–1.0 at 30–35 s. Although its BA is 0.42, QUANFRAUD uniquely combines GHZ/θ verification, DIDs, kernel privacy, audit trail, and calibration, prioritizing verifiability and security, establishing superior trust

TABLE XIII
ATTRIBUTE-WISE COMPARISON ACROSS REMAINING SCHEMES

Scheme	Trust	Kernel θ		DID	Audit	Calib.	Time (s)
QuanFraud (proposed)	**5.00**	✓	✓	✓	✓	✓	22
Q–LOF [16]	2.00	✗	✗	✗	✗	✓	30
RNN–BiLSTM [12]	1.00	✗	✗	✗	✗	✗	35
Quantum Banking [10]	2.00	✗	✗	✗	✓	✗	35

Note: ✓ indicates presence of the feature; ✗ indicates absence. "Kernel" = kernel privacy; θ = GHZ–θ protocol; "Calib.': probability calibration.

versus time and BA trade-offs. Uniquely, QUANFRAUD dominates the high-trust (5) region against time cost, despite modest BA under NISQ constraints, as shown in Fig. 9 and detailed steps are described with necessary coding that are linked and uploaded in the following GitHub Link: https://github.com/suman2025-kiit/Trust-Score-Calculation.git.

Accordingly, comparisons adopt an assurance-first lens: even when BA is modest, the conjunction of GHZ–θ [23], DID, kernel privacy, calibration, and on-chain auditability yields verifiability that competing schemes currently lack.

V. CONCLUSION AND EXTENSION

This paper introduces *QuanFraud*, a framework that pioneers the use of quantum state verification and blockchain to create a verifiable fraud-detection pipeline for IoT. Although the current implementation using QSVC demonstrates a significant trade-off between verifiability and detection accuracy (42.1% BA), it establishes a critical foundation for trust by coupling

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GHZ–θ verification, DID-anchored identity, kernel space privacy, and on-chain auditability. Our results show that classical detectors can exceed QSVC in discrimination. However, they do not provide state attestation, provenance, or tamper-evident logs, capabilities indispensable in adversarial IoT finance. Hence, *QuanFraud* reframes success from "best single metric" to "assurance-first" operation, where measurable trust, governance, and reproducibility are first-class results.

Our future work will focus on QML accuracy without sacrificing verifiability by exploring stronger quantum neural architectures, richer feature embeddings and re-uploading, and tuned hybrid quantum–classical models with calibrated thresholds. We will build a hardware prototype to measure end-to-end latency/throughput across GHZ generation, QSVC inference, and blockchain logging. We will formalize adversary and communication assumptions and prove security: quantum state verification soundness, identity binding, audit-trail integrity, resilience to leakage, as well as drift.

Code Availability

The authors have uploaded source code for our scheme with detailed replication steps and execution guidance available at the GitHub repository linked in Figs. 2(b) to 7,[3,4] and also here in the manuscript.[5]

References

[1] M. Yao, D. Tao, P. Qi, and R. Gao, "Scalable large model for unlabeled anomaly detection with trio-attention U-transformer and manifold-learning siamese discriminator," *IEEE Trans. Services Comput.*, vol. 18, no. 2, pp. 1012–1025, Mar./Apr. 2025.

[2] R. Lian, Y. Zheng, Y. Ming, C. Cai, C. Wang, and X. Jia, "Combating abusive information in encrypted messaging services: A secure and efficient realization," *IEEE Trans. Services Comput.*, to be published, doi: 10.1109/TSC.2025.3616516.

[3] P. Bhattacharya, P. Mehta, S. Tanwar, M. S. Obaidat, and K.-F. Hsiao, "HeaL: A blockchain-envisioned signcryption scheme for healthcare IoT ecosystems," in *Proc. Int. Conf. Commun., Comput., Cybersecurity, Informat.*, Sharjah, United Arab Emirates, 2020, pp. 1–6.

[4] H. M. Mujlid and R. Alshahrani, "Quantum-driven security evolution in IoT: AI-powered cryptography and anomaly detection," *J. Supercomputing*, vol. 81, no. 9, pp. 1–29, 2025.

[5] M. M. Saeed, "An AI-driven cybersecurity framework for IoT: Integrating LSTM-based anomaly detection, reinforcement learning, and post-quantum encryption," *IEEE Access*, vol. 13, pp. 104027–104036, 2025.

[6] S. Cherbal, A. Zier, S. Hebal, L. Louail, and B. Annane, "Security in Internet of Things: A review on approaches based on blockchain, machine learning, cryptography, and quantum computing," *J. Supercomputing*, vol. 80, no. 3, pp. 3738–3816, 2024.

[7] S. Tanwar, A. Popat, P. Bhattacharya, R. Gupta, and N. Kumar, "A taxonomy of energy optimization techniques for smart cities: Architecture and future directions," *Expert Systems*, vol. 39, no. 5, p. e12703, 2022.

[8] H. Gharavi, J. Granjal, and E. Monteiro, "Post-quantum blockchain security for the Internet of Things: Survey and research directions," *IEEE Commun. Surv. Tut.*, vol. 26, no. 3, pp. 1748–1774, Third Quarter 2024.

[9] A. S. Rajawat, S. Goyal, P. Bedi, T. Jan, M. Whaiduzzaman, and M. Prasad, "Quantum machine learning for security assessment in the Internet of Medical Things (IoMT)," *Future Internet*, vol. 15, no. 8, 2023, Art. no. 271.

[10] J. Mancilla and C. Pere, "Apreprocessing perspective for quantum machine learning classification advantage using NISQ algorithms," 2022, arXiv: *2208.13251*.

[11] D. Maheshwari, D. Sierra-Sosa, and B. Garcia-Zapirain, "Variational quantum classifier for binary classification: Real vs synthetic dataset," *IEEE Access*, vol. 10, pp. 3705–3715, 2022.

[12] E. Safeer, S. Tahir, A. Nawaz, M. Humayun, M. Shaheen, and M. Khan, "Advanced hybrid malware identification framework for the Internet of Medical Things, driven by deep learning," *Secur. Privacy*, vol. 8, 2024, Art. no. e454.

[13] H. Wang, W. Wang, Y. Liu, and B. Alidaee, "Integrating machine learning algorithms with quantum annealing solvers for online fraud detection," *IEEE Access*, vol. 10, pp. 75908–75917, 2022.

[14] S. K. Devineni, S. Kathiriya, and A. Shende, "Machine learning-powered anomaly detection: Enhancing data security and integrity," *J. Artif. Intell. Cloud Comput.*, vol. 184, pp. 2–9, 2023, doi: 10.47363/JAICC/2023(2).

[15] M. Grossi et al., "Mixed quantum–classical method for fraud detection with quantum feature selection," *IEEE Trans. Quantum Eng.*, vol. 3, 2022, Art. no. 3102812.

[16] M. Guo et al., "Quantum algorithm for unsupervised anomaly detection," *Physica A: Stat. Mechan. Appl.*, vol. 625, 2023, Art. no. 129018.

[17] D. Rajan and M. Visser, "Quantum blockchain using entanglement in time," *Quantum Rep.*, vol. 1, no. 1, pp. 3–11, 2019.

[18] H. Yang and M. Xiao, "Multi-user quantum private query," *Quantum Inf. Process.*, vol. 19, no. 8, 2020, Art. no. 253.

[19] X. Sun, M. Sopek, Q. Wang, and P. Kulicki, "Towards quantum-secured permissioned blockchain: Signature, consensus, and logic," *Entropy*, vol. 21, no. 9, 2019, Art. no. 887.

[20] T. M. Fernandez-Carames and P. Fraga-Lamas, "Towards post-quantum blockchain: A review on blockchain cryptography resistant to quantum computing attacks," *IEEE Access*, vol. 8, pp. 21091–21116, 2020.

[21] N. Mohanty, B. K. Behera, C. Ferrie, and P. Dash, "A quantum approach to synthetic minority oversampling technique (SMOTE)," *Quantum Mach. Intell.*, vol. 7, no. 1, 2025, Art. no. 38.

[22] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, 2002.

[23] A. Deshpande and M. Herlihy, "Privacy-preserving cross-chain atomic swaps," in *Proc. Int. Conf. Financial Cryptogr. Data Secur.*, Springer, 2020, pp. 540–549.

[24] M. S. Peelam and V. Chamola, "Enhancing security using quantum blockchain in consumer IoT networks," *IEEE Trans. Consumer Electron.*, 2024.

[25] D. Sierra-Sosa, M. Telahun, and A. Elmaghraby, "Tensorflow Quantum: Impacts of quantum state preparation on quantum machine learning performance," *IEEE Access*, vol. 8, pp. 215246–215255, 2020.

[26] S. Chakraborty, S. H. Shaikh, A. Chakrabarti, and R. Ghosh, "A hybrid quantum feature selection algorithm using a quantum inspired graph theoretic approach," *Appl. Intell.*, vol. 50, no. 6, pp. 1775–1793, 2020.

[27] S. Varma and R. Simon, "Bias in error estimation when using cross-validation for model selection," *BMC Bioinf.*, vol. 7, no. 1, 2006, Art. no. 91.

[28] G. C. Cawley and N. L. C. Talbot, "On over-fitting in model selection and subsequent selection bias in performance evaluation," *J. Mach. Learn. Res.*, vol. 11, pp. 2079–2107, 2010.

[29] H. Peng, F. Long, and C. Ding, "Feature selection based on mutual information: Criteria of max-dependency, max-relevance, and min-redundancy," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 8, pp. 1226–1238, Aug. 2005.

[30] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLoS One*, vol. 10, no. 3, 2015, Art. no. e0118432.

[31] Y. Ucbas, A. Eleyan, M. Hammoudeh, and M. Alohaly, "Performance and scalability analysis of ethereum and hyperledger fabric," *IEEE Access*, vol. 11, pp. 67156–67167, 2023.

[32] R. Ubale, K. K. Sujan, S. Deshpande, and G. T. Byrd, "Toward practical quantum machine learning: A novel hybrid quantum LSTM for fraud detection," 2025, *arXiv:2505.00137*.

[33] D. Shah and A. Sharma, "Benchmarking machine learning techniques for credit card fraud detection," *Indian J. Sci. Technol.*, vol. 18, pp. 1681–1695, 2025.

[34] D. P. Suda, G. Caruana, and S. Grima, "Comparison of tree-based learning methods for fraud detection in motor insurance," in *Proc. 14th Int. Conf. Data Sci., Technol. Appl.*, 2025, pp. 390–397.

[35] P. Komsrimorakot and T. Siriborvornratanakul, "Enhancing fraud detection in imbalanced motor insurance datasets using CP-SMOTE and random under-sampling," *J. Big Data*, vol. 12, 2025, Art. no. 172.

[3]https://github.com/drsumanmajumder/quantum-blockchain-.git
[4]https://github.com/suman2025-kiit/10-fold_validation__SMOTE.git
[5]https://github.com/suman2025-kiit/Trust-Score-Calculation.git

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